

E-COMMERCE DATA ENGINEERING & AGENTIC AI PLATFORM

Final Project Evaluation Report

Project Objective

This project demonstrates how Data Engineering, Business Intelligence, Machine Learning, and Agentic AI can work together to solve real E-Commerce business problems. Raw business data is processed into trusted analytical data, visualized through a dashboard, and then exposed through an AI chatbot that understands natural-language questions.

This project is an **end-to-end E-Commerce Data Engineering and Agentic AI system** that converts raw e-commerce data into meaningful business insights. It uses **ETL pipelines and Medallion Architecture (Bronze, Silver, Gold)** for data processing, **SCD Type 1 & Type 2** for handling historical data, and **OLAP/Power BI dashboards** for sales, profit, and forecasting analysis.

An **Agentic AI chatbot** is integrated with multiple specialized agents such as **Data Agent, ML/Forecast Agent, Insight Agent, Action Agent, Report Agent, and Pipeline Agent**. Users can ask questions in natural language, and the system analyses the data, provides insights, recommends actions, and generates reports.

In short:

Raw Data → ETL → Data Warehouse → Dashboard → Agentic AI → Insights → Recommendations.

Application: <http://localhost:8501>

Airflow: <http://localhost:8080>

1. Developer Introduction & Project Overview

I built this project to demonstrate how modern Data Engineering and Artificial Intelligence can be integrated into one practical analytics platform. The use case is an E-Commerce business where thousands of orders, customers, products, and transactions can generate large volumes of information.

In a traditional setup, a manager may need a data engineer or analyst to write SQL, search spreadsheets, prepare reports, and interpret trends. This project reduces that dependency by creating an automated data pipeline, an analytical dashboard, and a conversational AI layer.

Typical business questions:

- What are total sales this month?
- Which products are selling the most?
- Which region is most profitable?
- Why did profit decrease?
- What will sales look like next month?
- What action should management take?

Solution: The pipeline prepares reliable data, the dashboard presents it visually, and the Agentic AI layer allows users to ask questions in normal English and receive data-backed answers, charts, insights, and recommendations.

2. End-to-End Architecture

Source Data → ETL → Bronze → Silver → Gold → OLAP/Analytical Schema → BI Dashboard → Agent Controller → Specialized Agents → Data/ML/Report/Pipeline Tools → Validation → Insight/Action

Each layer has a clear purpose. Data Engineering makes the information reliable; the analyst layer turns it into a report; Agentic AI makes the platform conversational and intelligent.

3. Data Engineering Pipeline

The project uses Medallion Architecture. The easiest analogy is refining raw material step by step until it becomes ready for business use.

Stage	Layer	What happens
1	Bronze – Raw	Original source data is preserved with minimal modification, providing a reliable raw reference.
2	Silver – Clean	Formats, duplicates, missing values, currency symbols, dates, and other quality issues are handled and data is standardized.
3	Gold – Analytics Ready	Clean data is organized into business-friendly analytical structures such as sales, customers, products, and dates.

This separation improves maintainability, debugging, auditing, and reuse for analytics and AI.

4. ETL – Extract, Transform, Load

- **Extract:** Read data from the available source systems/files.
- **Transform:** Clean, standardize, join, aggregate, and derive required business fields.
- **Load:** Store processed information in the correct data layer.
- **Audit:** Record execution and processing information for monitoring.

Apache Airflow can orchestrate the workflow so the pipeline runs as a controlled sequence rather than as disconnected manual scripts.

5. SCD Type 1 & Type 2

Slowly Changing Dimensions manage changes in descriptive business information over time.

Type	Method	Example	Purpose
SCD 1	Overwrite old value	Latest product/customer attribute replaces the previous value	Keep current state
SCD 2	Create a new version and retain old record	Customer moves to another state/region; old and new versions remain with validity dates	Preserve historical accuracy

6. Data Quality & Protection

- **Identifier validation:** Invalid Order ID, Customer ID, or Product ID records are rejected or flagged according to pipeline rules.
- **Value validation:** Item prices and shipping charges are checked for valid positive values.
- **Duplicate/format checks:** Data is standardized and duplicate or malformed records are handled.
- **Audit logging:** Pipeline runs record records processed, passed, and failed.
- **SQL protection:** AI database access is restricted to safe read-only analytical operations.

Reliable AI depends on reliable data. These checks reduce the chance that bad input produces misleading analytics.

7. Schema-Based OLAP / Analytical Layer

After transformation, the project organizes information into structured analytical tables suitable for reporting and OLAP-style analysis. Sales, products, customers, dates, and related business entities can be queried through relationships and aggregations.

This is important for Agentic AI because the chatbot works with structured, cleaned, analytics-ready data instead of directly trying to understand raw files.

8. Data Analyst / BI Dashboard

The completed analyst component contains one report with three pages: **Sales, Profit, and Forecast**. It also includes the specified interaction and reporting features.

Feature	Implementation
Pages	3 – Sales, Profit, Forecast
Header / Hyperlink	1 header and 1 hyperlink
Charts	Required visualizations including line, bar, and pie charts
Prompt / Slicer	1 prompt and 1 slicer
Synced Slicer	1 synchronized slicer
Drill Operations	3 drill operations
Alerts	2 alert mechanisms
Page Navigation	2 navigation operations

9. Agentic AI – Multi-Agent Architecture

The AI layer is designed as a team of specialized agents. Instead of one chatbot trying to perform every task, an Auto Router/Controller identifies the user's intent and selects the required specialist agent or combination of agents.

Agent	Responsibility
■ Auto Router	Understands intent and routes the question to the correct specialist agent(s).
■ Data Agent	Uses schema information to generate and execute safe read-only SQL for sales, revenue, orders, products, customers, regions, and other supported metrics.
■ ML / Trend Agent	Calculates trends, moving averages, changes, spikes, dips, and other supported statistical patterns.
■ Forecast Agent	Uses the existing forecasting/ML implementation to estimate future sales or other supported metrics.
■ Insight Agent	Explains important patterns and possible contributors using retrieved evidence.
■ Action Agent	Converts supported insights into practical business recommendations.
■ Report Agent	Summarizes the business dashboard and provides an executive overview.
■ Pipeline Agent	Reports on ETL runs, audit information, and data-quality status.
■ General Agent	Handles supported questions about the project's technology stack and platform.

10. How a User Question Is Processed

User Question → Intent Detection → Agent Routing → Tool/Data Access → Validation → Insight Generation → Action (if required) → Final Response

For example, a simple sales question may use only the Data Agent. A complex question such as “Why did profit decrease and what should we do?” can involve Data Agent □ Insight Agent □ Action Agent.

The UI may show short operational statuses such as “Using Data Agent”, “Querying sales data”, “Running forecast”, and “Generating business insight”. Hidden chain-of-thought is not exposed.

11. Detailed Example – Profit Decrease + Recommended Action

User: “Why did profit decrease and what should we do?”

Step 1 – Auto Router: Detects that the request needs data analysis, explanation, and recommendation. It selects Data Agent □ Insight Agent □ Action Agent.

Step 2 – Data Agent: Reads schema metadata, generates a safe read-only query, and retrieves relevant profit/sales information from the analytical layer.

Step 3 – Insight Agent: Compares relevant periods, categories, products, or regions and explains the observed change using the retrieved evidence.

Step 4 – Action Agent: Converts the supported finding into a practical recommendation for management.

Step 5 – Final Response: The chatbot presents the answer, supporting numbers/chart where useful, the insight, and the recommended action.

Important: Numerical values must come from the project's actual database/model/report data. The AI must never invent a figure to make an answer look complete.

12. Fully Functional AI Chatbot

- **Natural-language input:** Ask questions in normal English.
- **Suggested prompts:** One-click questions such as Total Sales, Top 10 Products, and Sales Forecast.
- **Conversation context:** Follow-up questions can refer to the previous question.
- **Live status:** Display AI, database, and pipeline connection states where supported.
- **Visual results:** Show an appropriate line/bar/pie chart when numerical analysis benefits from visualization.
- **Tables:** Show detailed query results in a clean tabular format.
- **Error handling:** Unsupported questions, empty results, database errors, and model failures should be handled without crashing the application.

The chatbot is integrated into the existing application rather than being a separate static demo. Supported answers are generated dynamically from project data and models.

13. Typical Questions Supported

- What are total sales?
- What is total profit?
- Which product has the highest sales?
- Which region is most profitable?
- Show monthly sales.
- Show the top 10 products.
- What is the average order value?
- Compare sales between two years.
- Why did profit decrease?
- What are the major sales trends?
- What will sales be next month?
- What is the forecast for the next quarter?
- Which category should management focus on?
- What actions can improve profit?

14. AI Accuracy, Safety & Hallucination Prevention

A business analytics chatbot must be grounded in trusted data. The AI therefore follows a data-first approach.

- **No fabricated numbers:** Sales, profit, order counts, and forecasts must come from the database, analytical layer, report, or approved ML model.
- **Schema-aware SQL:** The Data Agent uses table and column metadata before generating queries.
- **Read-only access:** Destructive commands such as DROP, DELETE, UPDATE, INSERT, ALTER, and TRUNCATE are blocked for user-driven analytical requests.
- **Secret protection:** API keys and database credentials are stored through environment variables, not hardcoded source code.
- **Uncertainty handling:** If information is unavailable, the system clearly says it cannot answer from the available data.
- **Validation:** Generated SQL and returned results are checked before producing the final response.

15. Internal Structured Agent Communication

Agents can exchange compact structured information. For example, a Data Agent can return the detected intent, safe SQL, query result, and execution status; an Insight Agent can then use the verified result to produce an evidence-based explanation.

The user normally sees a clean business response rather than internal JSON or hidden reasoning.

16. Why This Qualifies as Agentic AI

- **Autonomous routing:** The controller chooses the relevant specialist.
- **Tool use:** Agents can use database/schema, forecast, report, and pipeline tools.
- **Multi-step processing:** Complex questions can require several agents.
- **Data grounding:** Results are tied to actual project data.
- **Business reasoning:** Results are converted into insights and actions.
- **Interactive experience:** Users can ask follow-up questions without manually writing SQL.

17. End-to-End Example Scenarios

Scenario A – Data

“What are total sales?” → Auto Router → Data Agent → Database → Validation → Answer

Scenario B – Forecast

“What will sales look like next month?” → Auto Router → Forecast Agent → Existing Forecast Model → Validation → Forecast Response

Scenario C – Insight

“Why are sales decreasing?” → Auto Router → Data Agent → Insight Agent → Evidence-based Explanation

Scenario D – Insight + Action

“Why did profit decrease and what should we do?” → Auto Router → Data Agent → Insight Agent → Action Agent → Final Recommendation

Scenario E – Pipeline

“Did the pipeline run successfully?” → Auto Router → Pipeline Agent → Airflow/Audit Information → Status Response

18. Technology Integration

- **Data Engineering:** ETL, Medallion Architecture, data validation, SCD Type 1, SCD Type 2, analytical/OLAP schema.
- **Orchestration:** Apache Airflow.
- **Database:** Existing project database and analytical tables.
- **Business Intelligence:** Existing three-page Sales, Profit, and Forecast report.
- **AI:** LLM-powered agent orchestration with specialized Data, ML/Trend, Forecast, Insight, Action, Report, Pipeline, and General capabilities.
- **Visualization:** Existing dashboard visuals and interactive charts, with Plotly where used.
- **Application:** Existing local web application at port 8501.

19. Key Features Demonstrated

- Multiple types of business data processed through a structured pipeline.
- ETL-based processing and transformation.
- Bronze, Silver, and Gold Medallion Architecture.
- SCD Type 1 and SCD Type 2 history management.
- Data-quality validation and audit logging.
- Schema-based analytical / OLAP-ready organization.
- Three-page Sales, Profit, and Forecast dashboard.
- Interactive filtering, synchronized filtering, drill operations, alerts, hyperlinks, and page navigation.
- Natural-language AI chatbot.
- Multi-agent routing and specialization.
- Safe SQL generation and execution.
- Trend analysis and forecasting.
- Business insight generation.
- Action/recommendation generation.
- Dashboard/report summarization.
- Conversation-aware follow-up questions.

20. Final Evaluation Summary

This project demonstrates a complete path from raw E-Commerce data to intelligent business decision support. The Data Engineering layer extracts, cleans, validates, transforms, historizes, and organizes data. The analyst layer converts that data into a business dashboard. The Agentic AI layer makes the platform conversational by allowing users to ask questions and receive data-backed answers.

The main achievement is the integration of these layers into one workflow: **Reliable Data → Analytical Insight → Conversational Access → Business Action.**

21. Conclusion

The E-Commerce Data Engineering & Agentic AI Platform demonstrates how modern data systems can be combined with AI to make analytics faster and easier to use. A manager does not need to know SQL, database structures, or complex analytical techniques to ask an important business question. The platform performs the technical processing in the background and returns an understandable result.

The project therefore demonstrates practical skills in Data Engineering, ETL, data quality, dimensional history management, analytical modeling, BI reporting, forecasting, LLM integration, multi-agent orchestration, safe tool use, and conversational analytics.

Final Architecture:

Source Data → ETL → Bronze → Silver → Gold → OLAP/Analytical Schema → BI Dashboard → Agent Controller → Specialized Agents → Database/ML/Report/Pipeline Tools → Validation → Insight → Action